pLog Book

Week 1: Tues

Immediate work to be done can be done:

- wheel drivers and chassis movement programming (STM)
- mechanical design decision
    I. open top with trash bag
    II. no ramp with pick up motion
    III. Yes ramp and scoop motion
    IV. camera placement
    V. skinny ramp (front side camera mount)

Discussion:

- Trash size benchmark, how big of trash to pick up
    I. Empty water bottle
    II. Dorito bag
    III. Jolly ranch wrapper?
- Shrink scope
    I. Hard set boundary test area, cut out need for lidar

Plan for next Thurs:

- Make mechanical decisions based on cardboard prototype

Plan Week 1 weekend:

- Chassis wheel actuation with drivers
- Stepper motor actuation

Goals for Winter quarter (said by McKell):

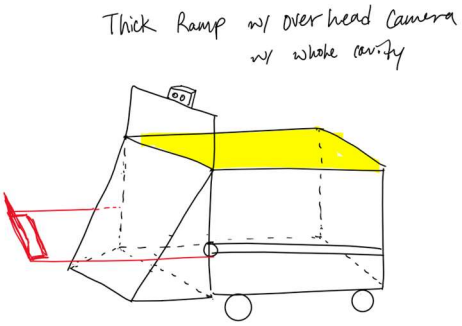
- 100% Design done
- 80% Purchasing done
- 50% Deliverables done
- 25% Testing done

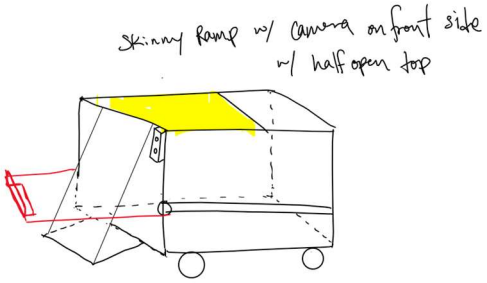
Week 1: Thurs

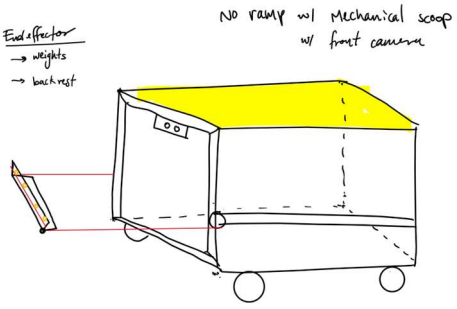
Objective: Make mechanical design decisions on lead screw height placement (DONE)

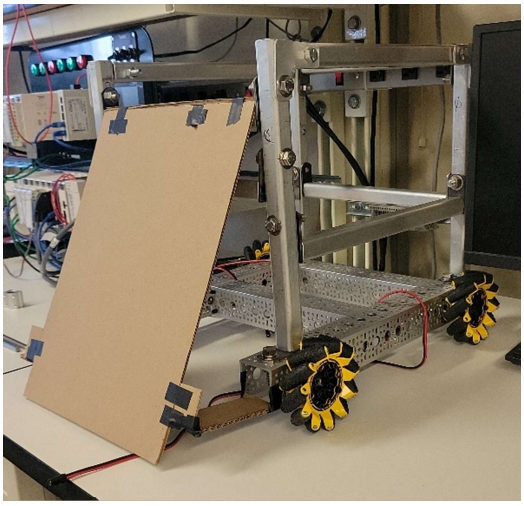
Progress:

- Lead screw (LS) height decided to be placed lower on frame
- LS and Linear slides will be attached to the bottom of middle frame
- Ramp prototype and design complete
    I. Fasteners on top where ramp meets top frame and where the bottom of ramp meets bottom strut
    II. CAD designs of ramp in progress for 3D print
- Camera placement ideas
    I. Above trash scoop radius, up front - like Wall-E (horizontal orientation), supports from the front of top frame
    II. On the side of the ramp closer to the ground (vertical orientation)
    III. Up front above arm radius with support from the back of the top frame
- Top front frame moved lower, creates shorter ramp and short arm. Less stress on servo









Week 1: Sat

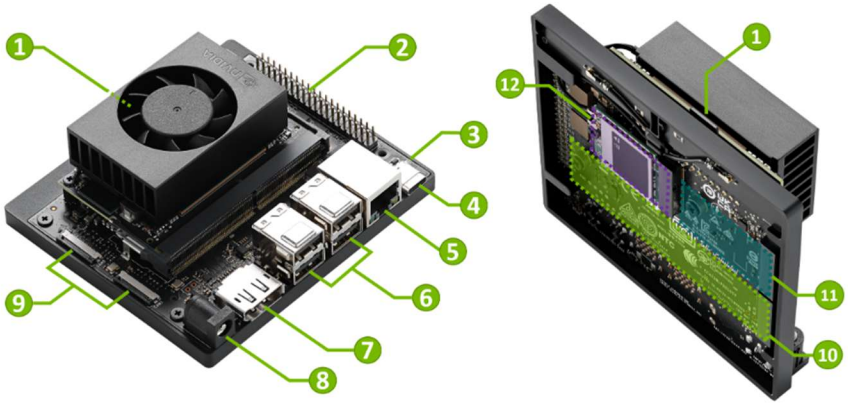
Objective: Chassis wheel activation, arm design

Progress:

- Chassis Wheel Activation
  I. One wheel successfully activated
  II. Using power supply (12VDC, 0.3ADC)
  III. Encoder hooked up to STM, printing motor shaft position
  IV. Driver successfully integrated
- Arm design
  I. One side of arm designed, 13 inch plus coupling
  II. End effector first draft designed, L shaped and boxy
- Mechanical structure update
  I. Drill hole marks for stepper and linear slides installation in progress

Week 2: Tues

Real sense Camera to Jetson Orin Board connection:



Part Names

Mark.	Name	Note
1	microSD card slot	
2	40-pin Expansion Header	
3	Power Indicator LED	
4	USB-C port	For data only
5	Gigabit Ethernet Port	
6	USB 3.2 Type-A ports (x4)	10Gpbs
7	DisplayPort Output Connector	
8	DC Power Jack	5.5mm x 2.5mm
9	MIPI CSI Camera Connectors (x2)	22pin, 0.5mm pitch
10	M.2 Slot (Key-M, Type 2280)	PCIe 3.0 x4
11	M.2 Slot (Key-M, Type 2230)	PCIe 3.0 x2
12	M.2 Slot (Key-E, Type 2230)	(populated)

1) What ports you're dealing with

- **Intel RealSense D455F connector: USB-C (USB 3.1 Gen 1)** on the camera. RealSense +2
- **Jetson Orin (common dev kits):**
  - **Jetson Orin Nano Dev Kit:** multiple **USB 3.2 Type-A (host) ports.** NVIDIA Developer
  - **Jetson AGX Orin Dev Kit:** has **USB Type-A host ports** and also Type-C ports, but the simple/typical approach is still "camera into a USB Type-A host port." NVIDIA Developer

2) The cable/adapter you should use

Best/most common

- **USB-C (camera) → USB-A (Jetson host)** using a **USB 3.x (SuperSpeed) cable.**
  Many D455 packages include a USB-C to USB-A SuperSpeed cable; if not, buy one that's explicitly USB 3.x rated. Amazon +1

Objective: Work on actuating motors = stepper, servo, chassis wheels

Progress: not complete

- Burnt 2 STM32 board, womp womp
- STM code written for servo, untested though (rotate 90 degrees and read position)
-

Week 2: Thurs

McKell advice: Make servo rotation a function of stepper position, so not to grind gears when

- Battery thoughts:

the wheels and scooping motion will not happen at the same time so we just need to see which operation pulls more current.

Initial thoughts is that the wheels would take more power, need to test with all the weight on

12V and 24V rails

Current spikes with Jetson connected with motors on the same supply (voltage, current regulator)

- If servo time sink, return smart and order simpler PWM servo

Stepper Pinout

- PUL+ → PA0
- PUL- → GND
- DIR + → PB0
- DIR- → GND
-

Week 2: Sat

Objectives:

- activate all four wheels, complete rudimentary turns.
- Establish communication between the Jetson and STM

Progress:

- All wheels were wired to theoretically working pins. Not tested yet. Did not achieve turning yet, push to Week 3 Thurs

- Terminal communication between STM and Jetson established. Packet of information from board to board unsuccessful

4 Wheel PINOUT : FL/FR/RL/RR

```
/* ========================
  Example pin mapping
  ========================
  PWM (TIM2): PA0/PA1/PA2/PA3 -> CH1..CH4
  DIR pins : PB0/PB1/PB2/PB10

  Encoders:
   W_FL: TIM3 on PA6/PA7 (CH1/CH2)
   W_FR: TIM4 on PB6/PB7
   W_RL: TIM1 on PA8/PA9
   W_RR: TIM8 on PC6/PC7
*/

Wheel_t g_wheels[4] = {
  // W_FL
  { TIM2, 1, GPIOB, GPIO_PIN_0,  true,  TIM3, 0, 0 },
  // W_FR
  { TIM2, 2, GPIOB, GPIO_PIN_1,  true,  TIM4, 0, 0 },
```

```
-       // W_RL
-       { TIM2, 3, GPIOB, GPIO_PIN_2,  true,  TIM1, 0, 0 },
-       // W_RR
-       { TIM2, 4, GPIOB, GPIO_PIN_10, true,  TIM8, 0, 0 },
-   };
```

Week 3: Thurs

1/22/2026

Jetson – STM32 COMMS

Capabilities:

- Keypress Movement:
  - Locomotion
  - Arm Sequence
- Data Collection (keypress or maybe not)
  - Lidar
  - Camera (RGB, Depth)
  - Serial COMMS:
    - Tx and Rx
  - Wheel Odometry
  - SLAM (Positional Data Relative to Current Location)
- GUI with Estop (subscriber to other slam nodes)

Communication Layer:

Mckell Talk:

- Research Jetson internal power regulation, could boost a 12V to 19V
- Power rails for 7 peripherals from battery could be simple wire to battery. So no additional rails just motor straight to battery terminals
- If need power level shifting for jetson (24V → 19V) then power board could be worth it

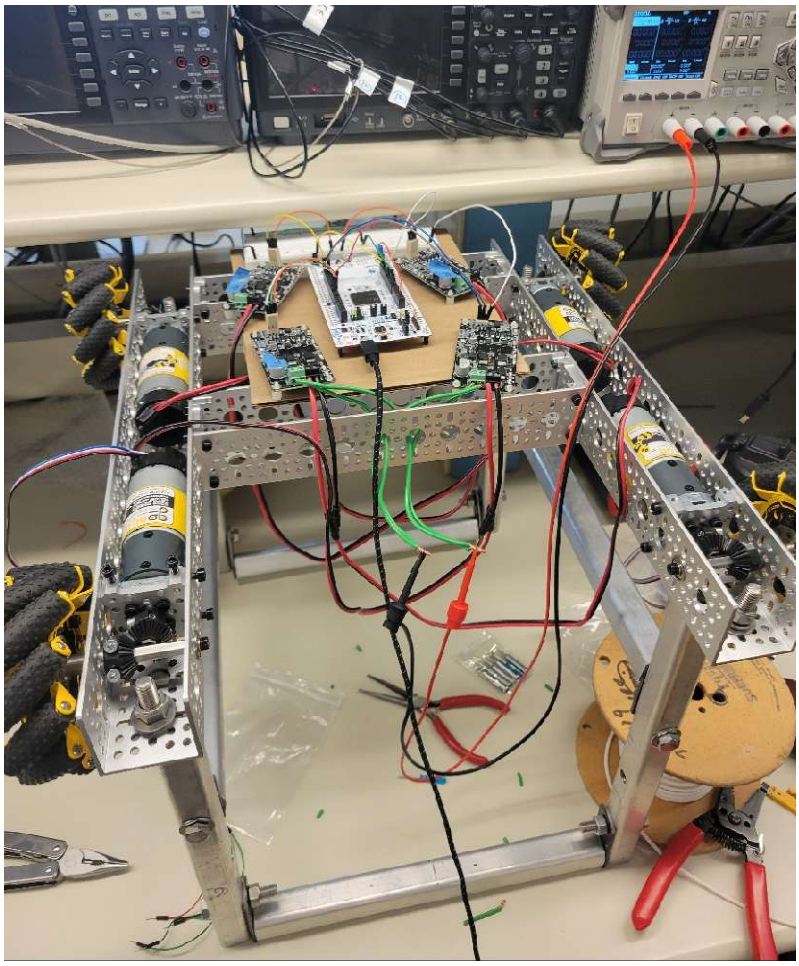
Progress:

- Packet communication between Jetson and STM to one wheel motor achieved (rotate CCW & CW)
- Github folders laid out, FIRMWARE = STM32 code (C), SOFTWARE = Jetson code (python)

Week 4: Tues

Progress:

- Actuated 4 wheels using keyboard commands → Jetson → STM32 → motors
    - Can move forward, reverse, and possibly strafe
    - Right now each direction actuation has a stutter effect, short spin – stop – spin for as long as actuated
    - Idle spinning current = 0.59 A (2 wheels, no additional weight)
    - Current spike when actuated or change direction = 1.8A (2 wheels, no additional weight)



4 wheel Motor Wiring Diagram:

FL (red)

- PWM → PA0
- DIR → PB0
- GND →
- CHA → PB6
- CHB → PB7

FR (blue)

- PWM → PA1
- DIR → PB1
- GND →
- CHA → PA6
- CHB → PA7

RL (green)

- PWM → PA2
- DIR → PB2
- CHA → PC6
- CHB → PC7

RR (yellow)

- PWM → PA3
- DIR → PF14
- CHA → PE9
- CHB → PE11

White is CHA, Blue CHB (encoders)

Stepper Motor

- DIR → PF15
- PWM → PD14

Servo Motor

- UART3 → PB10

soldered

Week 4: Thurs

Progress: Encoders hooked up to STM, updated wiring diagram listed above

McKell Talk: make battery selection more of a priority

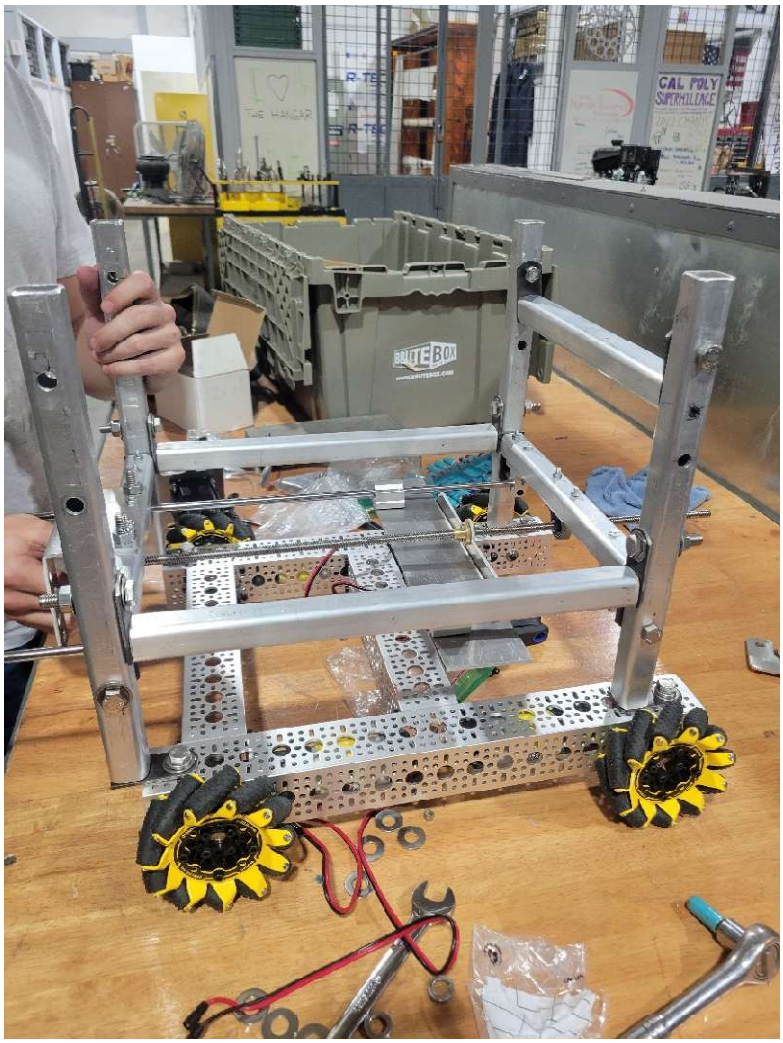
Next Steps:

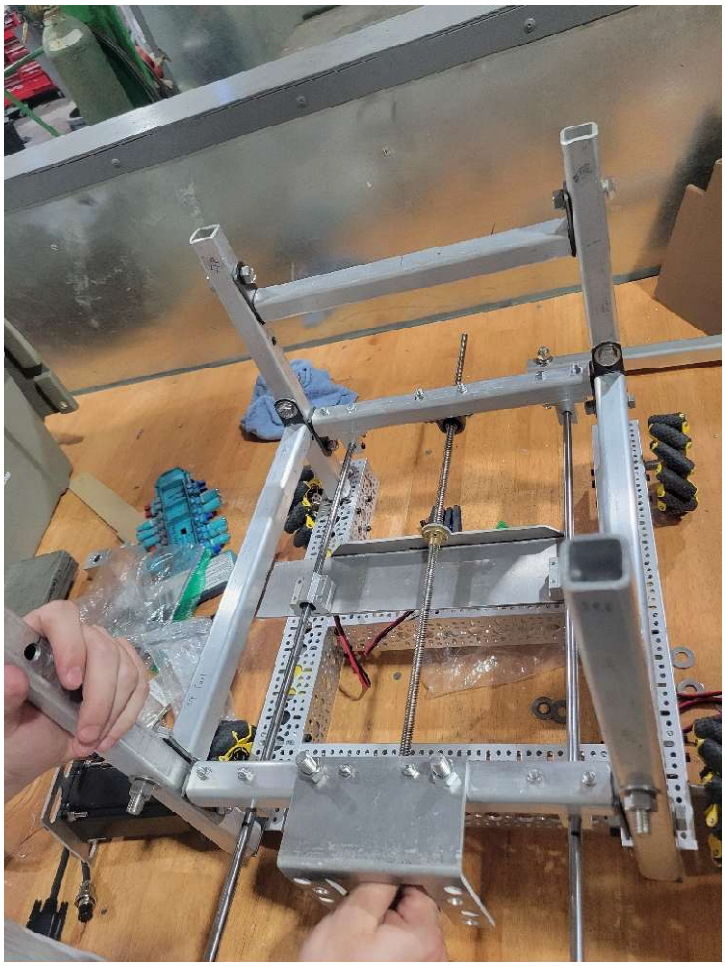
Mechanical assembly → space allocation → battery purchasing → scoop trajectory →

Week 5: Tues

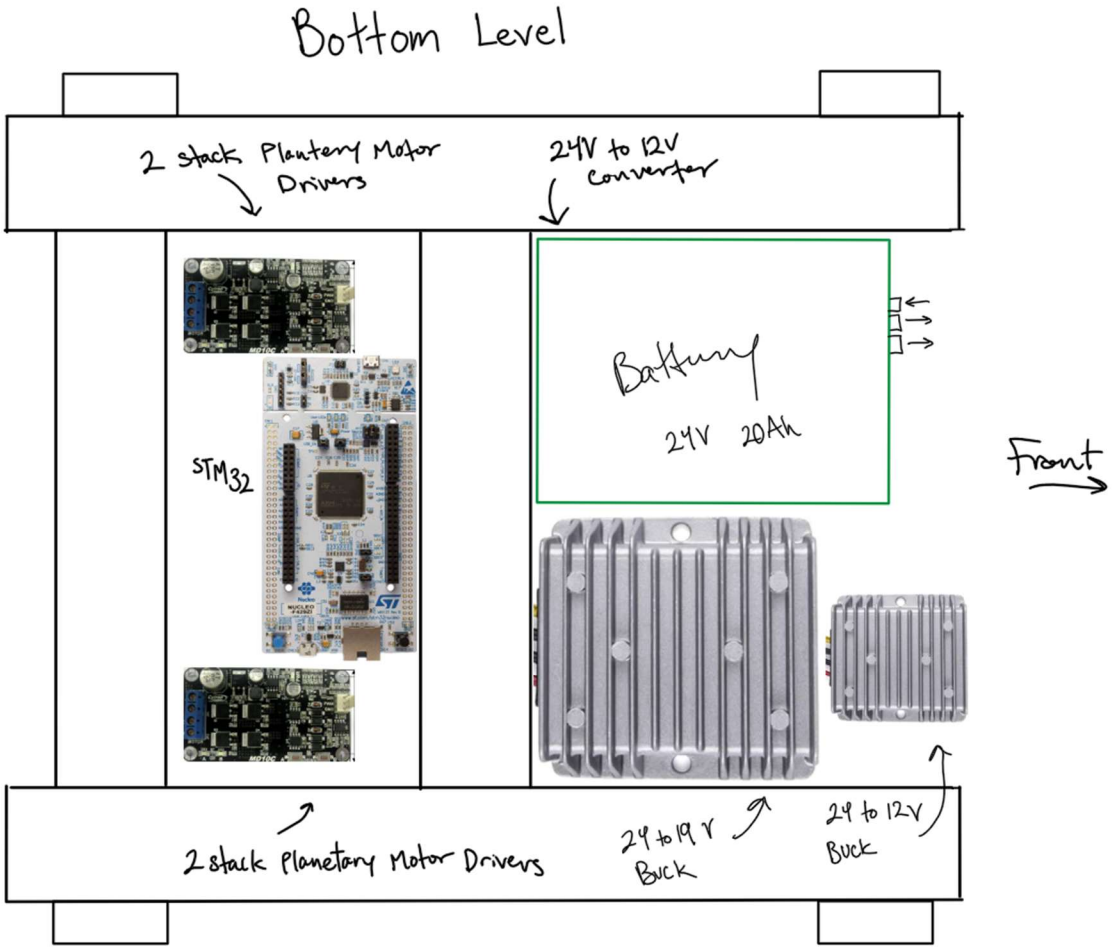
Progress: mechanically assembled linear slides and, electrical layout for bottom level designed

Linear Slides Pictures:


CAL POLY
SUPERMILEAGE



Preliminary Bottom Level electronic layout:

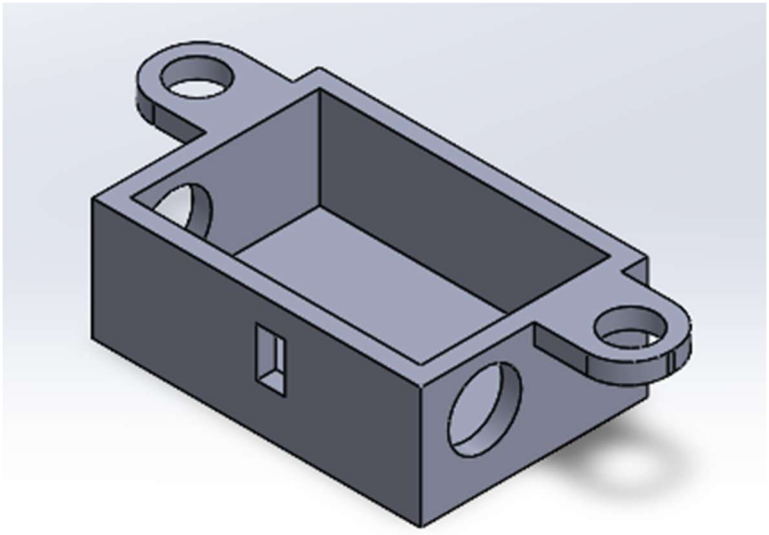


- STM and Driver location confirmed, housing printed (in the future) but not installed
- New, smaller battery bought. Height might be an issue for the bottom level, might hit servo housing when linearly actuated, have to test
- Buck converts can co in this front cavity, would have to create housing. They could also be located on second level near the jetson

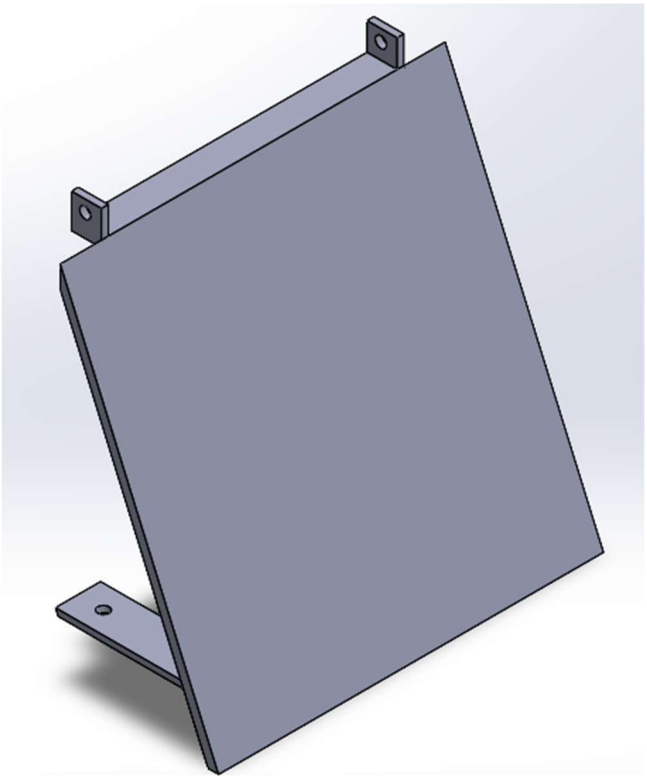
Week 5: Thurs

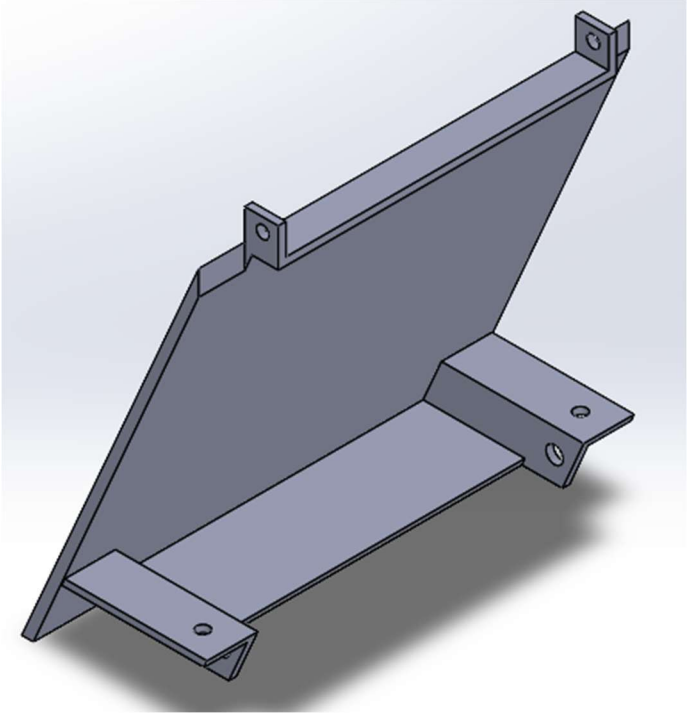
Progress: Servo mount designed and printed, RampV3 designed, STM and driver housing design began

Servo Mount:



RampV3:

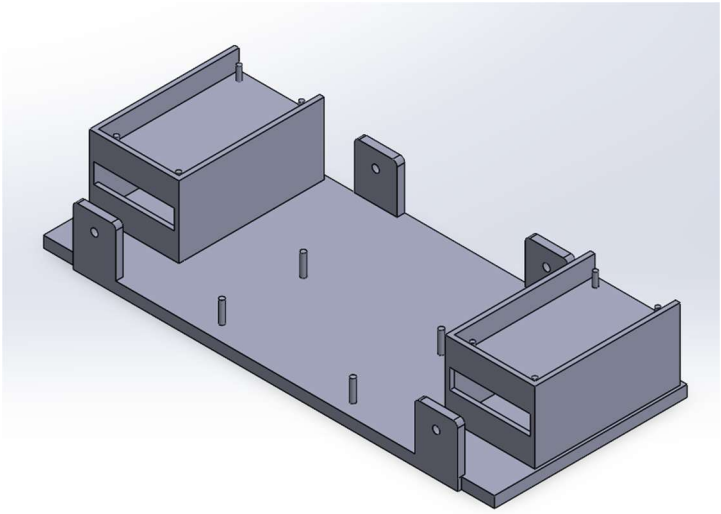


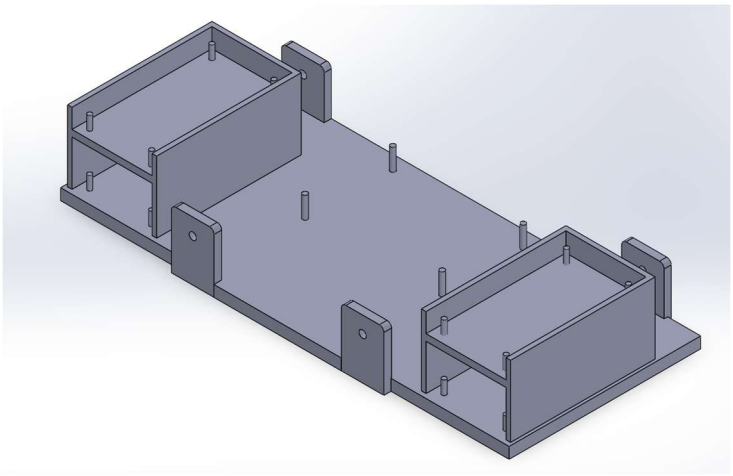


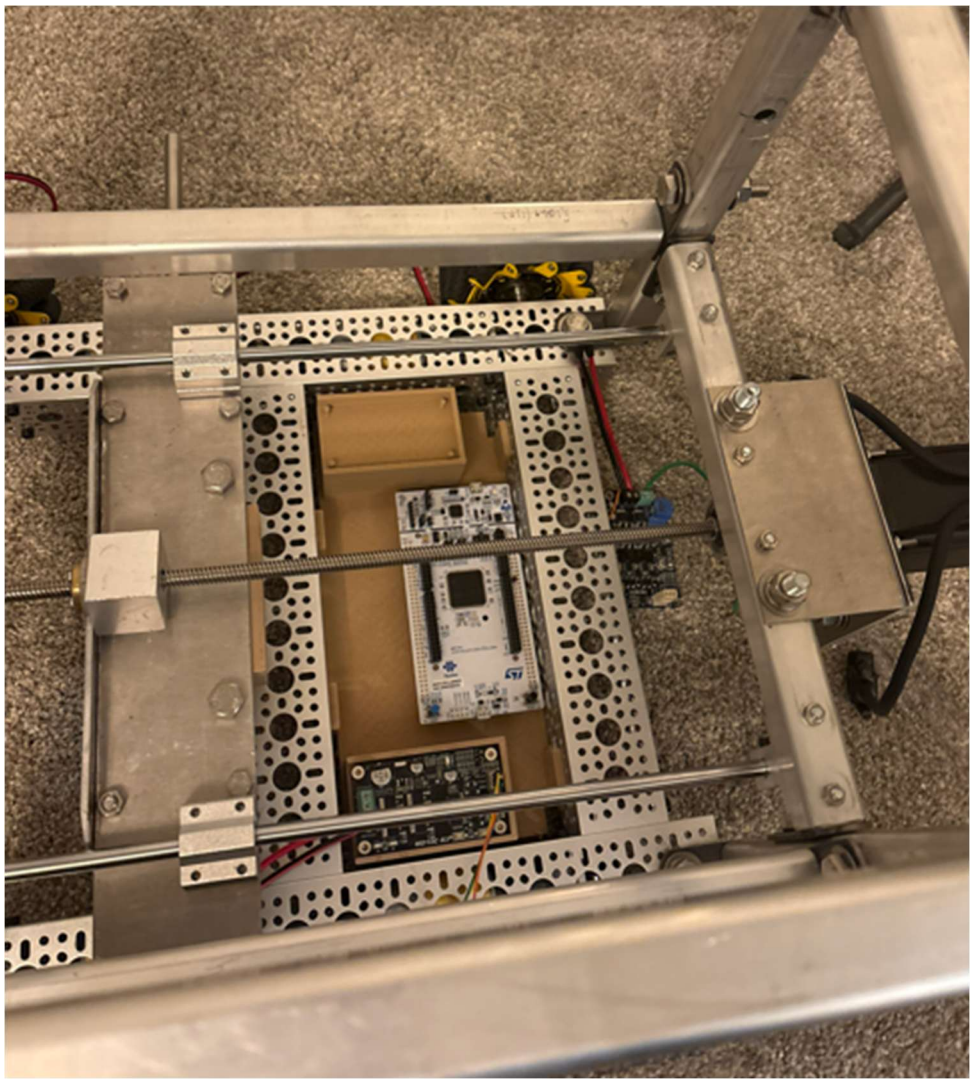
Week 6: Tues

Progress: Robotic arm assembly complete, RampV3 print began (Wed), STM and driver housing complete and printed

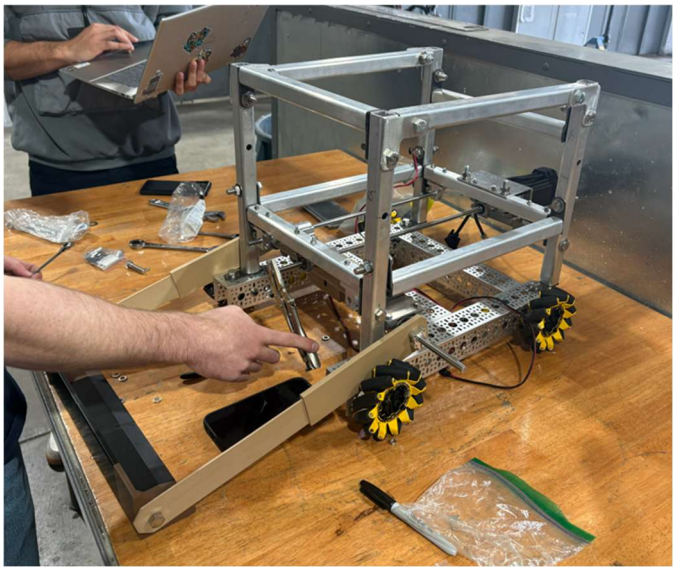
STM and driver housing:

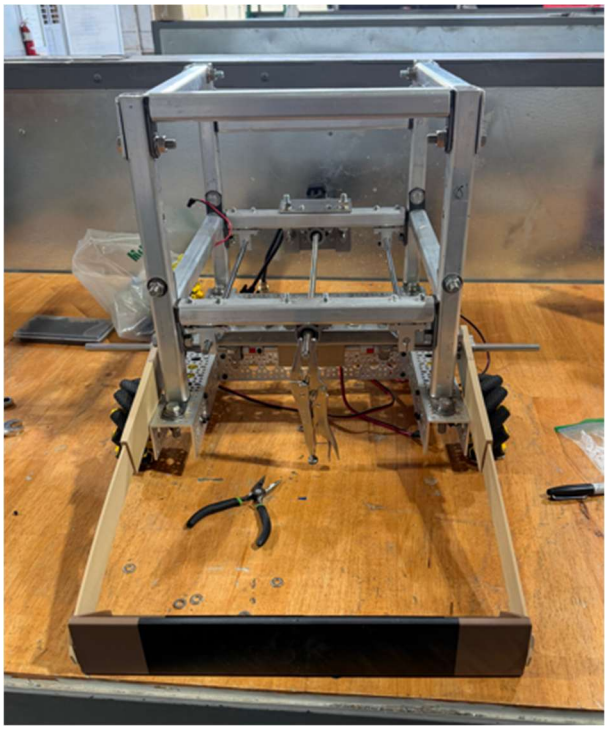






Robotic arm assembly:







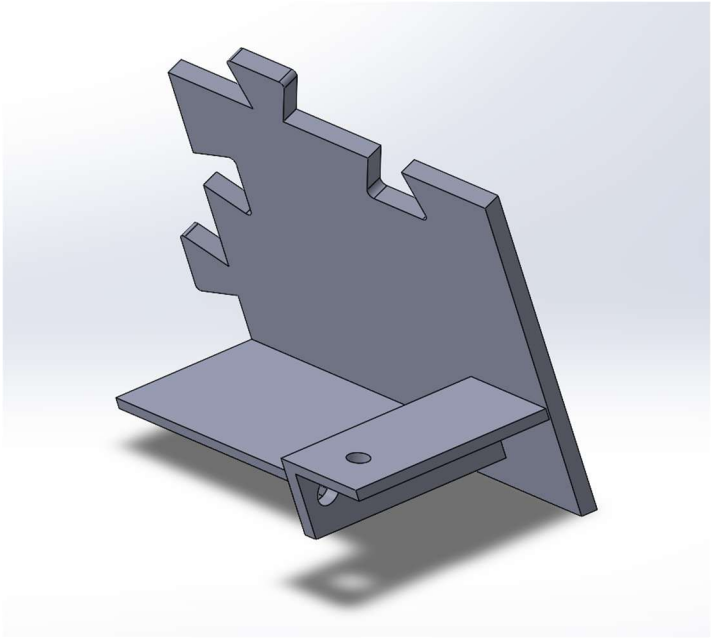
Week 6: Thurs

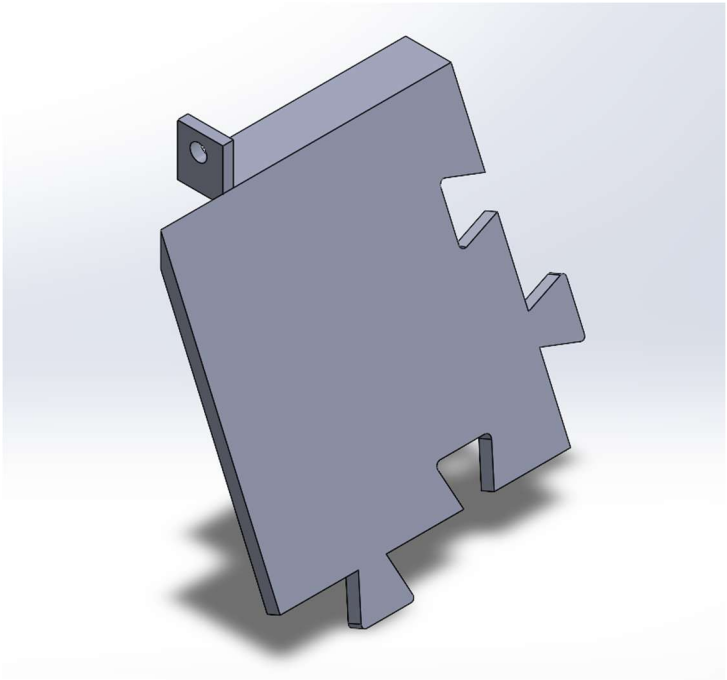
Progress: Finalized electronic layout, finalized puzzle piece ramp

Electronic Layout:

- Bottom Layer = battery in front cavity, STM in back cavity in 3D PLA print housing, drivers in the middle cross strut screwed into frame.
- Middle Layer = jetson board, stepper driver, both housed on leftover aluminum sheet
- Top layer = LiDAR, camera, and voltage converters housed on leftover sheet metal. LiDAR towards the back, camera towards the front (behind trash cavity), and voltage converts screwed on the under side of the top sheet

Puzzled Pieced Ramp





Week 7: Tues

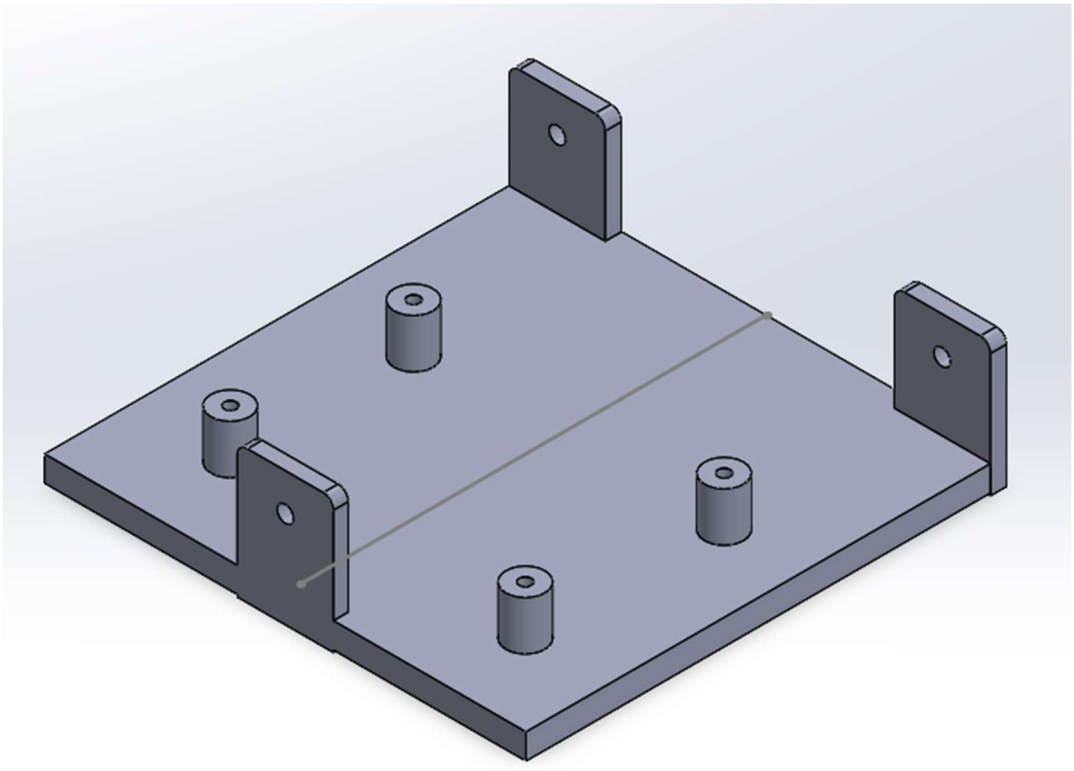
Progress: More mechanical assembly! Top and Middle layers have been assembled (sheets cut and peripherals attached). Bottom Layer assembled but STM housing needs to be iterated on. Battery has been installed. 3 out of 4 ramp pieces printed and assembled, waiting on 4th piece to successfully print.

Big next step is overall wiring.

Pictures of this assembly coming soon:

[Insert Pictures]

STM Housing Revised:



Week 8: Tuesday

This week we achieved full system mobility using ROS2 teleop. The robot successfully drove around under battery power with all major subsystems running together, including motor control and onboard electronics. This marked the first time the rover operated untethered in a fully integrated configuration.

During additional testing of the bus servo, we encountered a wiring issue that resulted in damage to the microcontroller. We are now preparing a replacement STM32 board and will rewire the servo system more carefully to prevent recurrence.

Key Takeaways:

- Full battery-powered integration is functional
- ROS2 teleoperation pipeline is working end-to-end
- Greater care is required when interfacing higher-voltage components with MCU logic
- Weight: 42 lbs

Next Steps:

- Rewire STM
- Test servo and stepper motors operating in full system
- Code full system test script including teleop locomotion and arm sequence command

Servo Procedure

- On ribbon on servo housing with curve side up left to right G, P, BUS
- On ribbon on STM side with curve side up left to right BUS, P, G
- Powered with GND to common GND through Jetson, STM shouldn't need a separate reference because powered with same GND
- Rotation against gravity 0.4A

STM Power

- Wheel encoders (3.3V) take about on the high end 20mA
- Wheel driver logic (negligible)

- Stepper driver 5V logic, can be powered by 3.3V tested. (negligible current), encoder is internally powered by stepper